



Sadguru Balumama Shikshan Prasarak Mandal's
K.P.PATIL INSTITUTE OF TECHNOLOGY

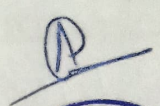
(DTE CODE : 6814) (MSBTE Code : 1661)

Approved by AICTE, DTE Mumbai & Govt. of Maharashtra, Affiliated to MSBTE Mumbai

Internships Daily Diary(2025-26)

Department Of Artificial Intelligence & Machine Learning Engineering

Program Code : **AI5K**
Name of the Student : **Sanika Dhanaji Chougale**
Name of the Mentor (Faculty) : **Mrs. P.M. Kumbhar**
Name of the Mentor (Industry): **Mr. Abhijit Gatade**
Enrollment Number : **23213500044**

Week	Day & Date	Discussion Topics/Activity	Details of Work Allotted Till Next Session /Corrections Suggested/Faculty Remarks	Signature of Industry Mentor
Week <u>9</u>	Mon, Date <u>28/07/25</u>	♦ Started Movie Recommendation Project	Studied dataset and filtering methods	 
	Tue, Date <u>29/07/25</u>	♦ Data cleaning and Preprocessing	Created tags, Column, cleaned text.	
	Wed, Date <u>30/07/25</u>	♦ Feature Extraction	Applied Countvectors & Similarity	
	Thu, Date <u>31/07/25</u>	♦ Logic Implementation	Wrote function to recommend movies	
	Fri, Date <u>1/08/25</u>	♦ Pickle file creation	Saved model & similarity files	
	Sat, Date <u>02/08/25</u>	♦ eamlit UI	Built UI and tested the system	



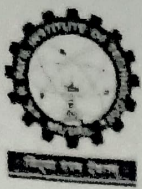
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Week No: 09	Day: Monday	Date: 28/07/25
<u>" Project Initiation and Dataset "</u>		
The internship began with initiating the movie Recommendation System project. I focused on understanding the problem Statement and identifying the goals of the project. The dataset was imported and explored in detail. I examined the Columns and types of the data available (like movie titles, genres, keywords, overview, cast, and crew). I also researched different filtering methods to analyze and understand how the system could suggest movie based on user input. This gave me a solid foundation to move forward with cleaning and processing the data.		



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Week No: 09

Day: Tuesday

Date: 23/07/25

"Data cleaning and Preprocessing"

On this day, I performed extensive data cleaning to prepare the dataset for model training. I handled missing values, removed unnecessary columns, and combined relevant columns like overview, genres, cast, crew, and keywords into a single 'tags' column to represent each movie in a unified text format.

The combined column was then cleaned by removing special characters, converting to lowercase and stemming the words to their base form. This preprocessing step was essential to ensure consistency and to extract meaningful feature in the next phase.



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Week No: 09

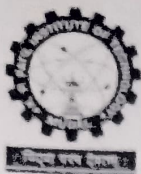
Day: Wednesday

Date: 30/07/25

" feature Extraction "

I implemented feature extraction using CountVectorizer, which transformed the text data into a matrix of token counts. Then, I applied cosine similarity to compute the similarity between movies based on their tags.

This allowed me to measure how closely two movies relate in terms of content. I analyzed similarity scores to identify the top recommendation for a given movie. This was a key step in developing the recommendation logic and required a good understanding of text vectorization and similarity measures in natural languages processing.



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Week No: 09

Day: Thursday

Date: 31/07/25

"Logic Implementation for Recommendation"

I wrote the main python function that powers the recommendation engine. The function takes a movie title as inputs, fetches its index from the dataset, retrieves the similarity scores, and sorts them to find the top N most similar movies. The function formed the core of the recommendation system. I tested it by giving various movie titles as input and validated that the recommendations were accurate and relevant. This task enhanced my programming skills and deepened my understanding of logic design and algorithm writing.



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Week No: 09

Day: Friday

Date: 01/08/25

"Pickle file Creation"

To make the project reusable and deployable, I saved the processed model and similarity matrix using python's pickle module. I created .pkl files for the movie dataset (movies - dict .pkl) and the similarity matrix (similarity .pkl)

This ensured that the model could be loaded in the future without repeating the entire processing steps. It was an important step for deployment and practical implementation of the project, especially in web-based system.



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Week No: 09

Day: Saturday

Date: 02/08/25

" User Interface Development "

I Created a simple User interface (UI) using streamlit, a python -based web framework.

The UI allowed users . to enter a movie name and get real-time recommendations. I integrated the backend logic and pickle files into the streamlit app, and tested the full system multiple times to ensure proper functionality. The system was able to recommend movies quickly and accurately, Completing the core objective of the project.